

Global Dysfunction (Encephalopathy)

Altered mentation (encephalopathy), typically is a consequence of impaired functioning of both cerebral hemispheres and/or the brainstem (global dysfunction). Worsening encephalopathy may result in coma which is a deep and prolonged state of unconsciousness characterized by unresponsiveness to external stimuli, absence of sleep-wake cycles, and the lack of any voluntary actions [1].

The EEG is exquisitely sensitive to global cerebral dysfunction but is poorly specific to etiology. Multiple etiologies of encephalopathy may lead to similar electrographic patterns but have different prognosis. These patterns are typically abnormalities of background (Chapter 10) and repetitive abnormalities of foreground (Chapter 15). Less commonly, repetitive patterns represent ictal activity that contributes to the encephalopathic state (nonconvulsive status epilepticus) [2].

Occasionally, an EEG may appear normal in comatose individuals suggesting psychogenic coma or a locked-in state [3]. A handful of “classic” EEG presentations exist that may suggest a specific cause of encephalopathy, these are described in the later part of this chapter.

The reader’s basic approach should be like that described for a focal lesion:

1. make diagnosis;
2. estimate severity;
3. identify any repetitive patterns and understand their implications;
4. identify nonconvulsive status epilepticus (NCSE).

Diagnosing Encephalopathy on EEG

Encephalopathy, which is synonymous with diffuse, global, or bilateral hemispheric dysfunction, is a common electrographic diagnosis in those with altered mentation. The three cardinal features of an encephalopathic EEG include (1) background slowing, (2) attenuation/discontinuity, and (3) decreased reactivity. In the initial stages of encephalopathy, there is slowing of the posterior-dominant rhythm (differentiate this from excessive drowsiness, where the PDR is brief but reactive and within alpha range), the background

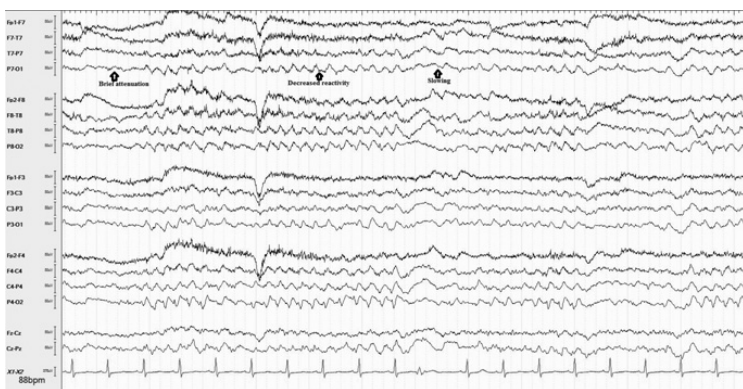


Figure 23.1 Fifty-two-year old man with confusion, early encephalopathic changes on EEG.

amplitude may be increased with emergence of theta and delta activities but later attenuates, and there is paradoxical or decreased reactivity to both endogenous state changes (such as arousals) or external stimulation. Paradoxical alpha refers to the emergence of an occipital alpha with eye opening and is typical of early encephalopathy [4].

Figure 23.1 shows early encephalopathy with emergence of diffuse slowing, brief periods of attenuation, and decreased reactivity of the background.

Estimating Severity

Progressive deterioration in the level of consciousness is typically associated with background slowing; there is emergence of theta and delta activities which may completely replace the normal background in more severe forms (Figure 23.2). Initially, there may be an increase in the background amplitude (voltage), but with worsening cerebral dysfunction, there is attenuation and then suppression of cerebral activity resulting in discontinuity. Burst suppression or even complete electrocerebral silence may be seen in the most severe forms. The progression of discontinuity is shown in Chapter 10, Figures 10.3, 10.4, and 10.5.

Loss of reactivity (and variability) is another important electrographic marker of severity. The loss of reactivity may be assessed by the technologist in a graded manner using increasing levels of stimulation. The clinical and electrographic response along with the specific stimulation should be recorded in real time.

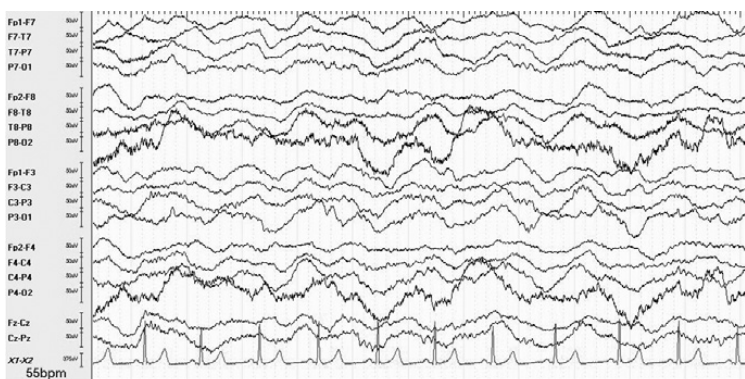


Figure 23.2 Same patient as in Figure 23.1, diffuse high-amplitude delta slowing with progression on encephalopathy.

Though background slowing, amplitude attenuation, and loss of reactivity are associated with increasing severity of encephalopathy, these features may not always occur in perfect sequence, nor do they precisely correlate with the bedside clinical examination. However, they collectively offer a useful overall assessment of the depth of encephalopathy, which may be trended over time using either follow-up EEGs or continuous monitoring. Reversal of these trends may indicate improvement.

Severe encephalopathies (such as traumatic brain injury or cerebral anoxia) are typically associated with slow, low-voltage, and unreactive electrographic patterns. However, an estimate of severity should not be confused with prognostication, as this is dependent on the etiology. A severe encephalopathy due to a reversible cause such as a medication-induced coma may have a similar electrographic appearance to anoxic brain injury (e.g., both may show burst suppression) but portend very different outcomes [5].

Identifying Repetitive Patterns and Understanding Their Implications

Generalized and lateralized repetitive abnormalities are commonly observed in encephalopathic patients; they're described in Chapter 15. Identification of these repetitive patterns has important implications regarding etiology, potential for epileptic seizures (epileptogenicity), neuronal injury, and prognosis. These points are discussed below.

Etiology: Though poorly specific, electrographic patterns are suggestive of the underlying etiological process. Generalized rhythmic delta activity (GRDA) and generalized periodic discharges with triphasic morphology (triphasic waves) are commonly observed in toxic-metabolic encephalopathies. Lateralized periodic discharges (LPDs) typically suggest an acute destructive process (such as herpes encephalitis), and generalized periodic discharges (GPDs) may be associated with nonconvulsive status epilepticus (NCSE).

Epileptogenicity: Common repetitive patterns that are strongly associated with epileptic seizures include lateralized rhythmic delta activity (LRDA), generalized periodic discharges (GPDs), bilateral periodic discharges (BIPDs), and lateralized periodic discharges (LPDs). Occipital intermittent rhythmic delta activity (OIRDA) may be associated with idiopathic generalized epilepsies, as shown in Figure 23.3.

Prognosis: Periodic patterns are electrographic markers of neuronal injury; consequently, they are associated with poor outcomes. However, it is not clearly understood if these patterns are merely the manifestation of neuronal injury or themselves have some causal role. Both GPDs and LPDs have been traditionally associated with increased mortality. Though recent studies do not show a direct association with increased mortality, GPDs are associated with NCSE, which itself is independently associated with increased mortality. Rarely, cyclical alternating patterns (CAP) may be seen in coma; these typically consist of alternating sequences of high-amplitude slowing with low-amplitude faster activity. They may represent state changes and are associated

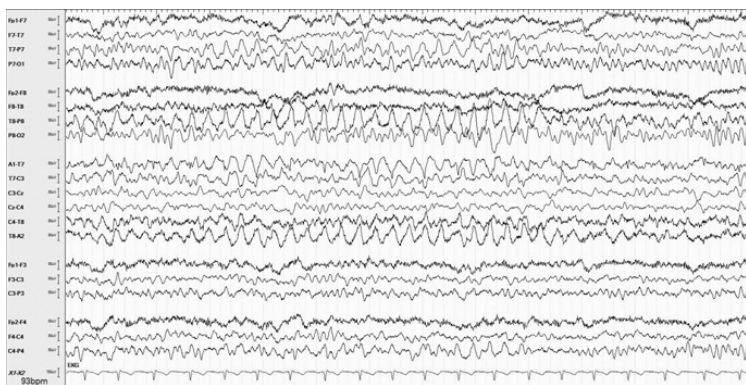


Figure 23.3 Eight-year-old boy with childhood absence epilepsy, runs of occipital intermittent rhythmic delta activity (OIRDA).

with favorable outcomes. However, it should be emphasized that etiology and not the electrographic pattern alone is the key determinant of clinical outcome [6,7].

Diagnosing Nonconvulsive Status Epilepticus

Rhythmic and periodic lie on an ictal–interictal continuum; at the ictal end they represent ongoing epileptic activity that is potentially contributing to the encephalopathic state without overt clinical signs – this is called nonconvulsive status epilepticus (NCSE). There are no universally accepted criteria for the diagnosis of NCSE. In practice, a diagnosis requires a combined assessment of clinical signs (altered mentation, aphasia, subtle motor, and behavioral signs) and EEG findings (continuous or recurrent electrographic ictal activity) that last 5 min or longer and may improve with a trial of antiepileptic medications. A diagnostic response to antiepileptics (such as lorazepam) must involve both clinical and electrographic improvement (resolution of ictal activity and crucially, return of normal background). These responses may be delayed or equivocal and not necessarily rapid, as described in Chapter 15. Timely identification of NCSE is important as it has been independently associated with increased mortality, and a diagnostic delay may contribute to worse clinical outcomes. NCSE is described in Chapter 24. Figure 23.4 shows NCSE.

Additionally, the reader must be familiar with a handful of “classical” electrographic presentations of encephalopathy with specific diagnostic

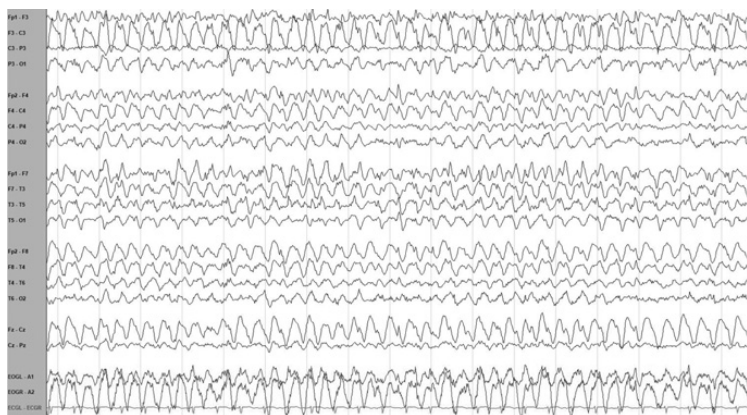


Figure 23.4 Twenty-seven-year-old woman with anti-NMDAR encephalitis, continuous generalized rhythmic discharges consistent with NCSE.

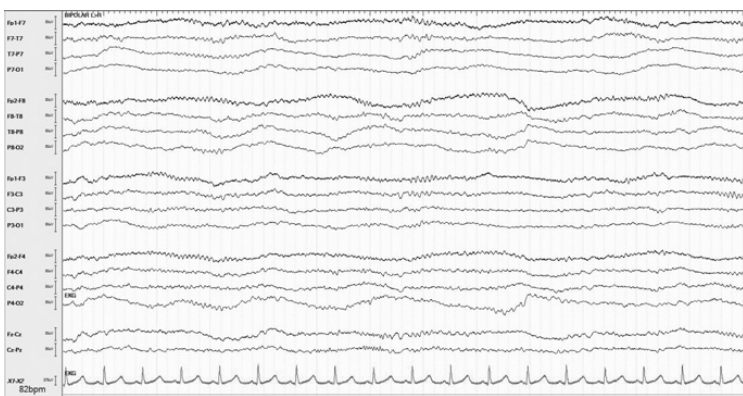


Figure 23.5 Thirty-six-year-old man with narcotic overdose, spindle coma pattern.

implications. These include spindle coma, alpha-theta coma, CJD, SSPE, and extreme delta brush.

Spindle Coma

The EEG is characterized by a diffuse theta and/or delta background with frequent 11- to 14-Hz symmetric spindles. On stimulation, K complexes may be seen along with other sleep architecture, such as vertex waves, however, clinically, the patient remains unconscious without arousals or fluctuations. Spindle coma is commonly associated with traumatic brain injury, drug overdoses, cerebral anoxia, and viral encephalitis. Anatomically, the pattern suggests involvement below the level of the thalamus, which is where spindles are thought to originate (such as Ponto mesencephalic junction). Prognosis is thought to be favorable depending on the presence of reactivity and absence of structural damage. If coma deepens, spindle frequencies may slow. Figure 23.5 shows a spindle coma pattern in the setting of a narcotic overdose [8].

Alpha/Alpha-Theta Coma

The EEG is characterized by mixed alpha and theta frequencies. There are a few distinct subtypes based on etiology:

1. Alpha coma associated with cerebral anoxia is characterized by diffuse or frontally predominant, monomorphic, unreactive alpha (continuous featureless alpha) with a grim prognosis (Chapter 25).

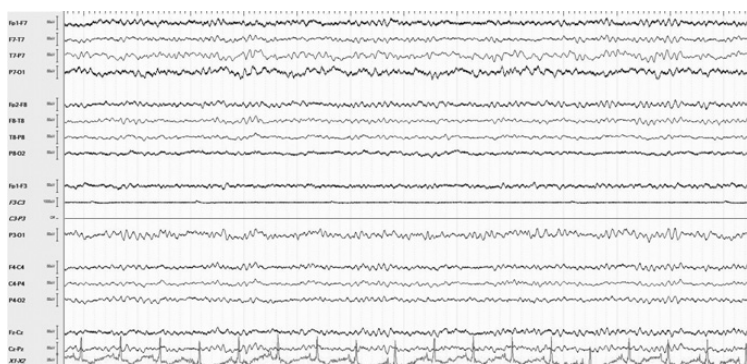


Figure 23.6 Forty-seven-year-old man comatose after pontine tumor removal, diffuse alpha activity (C3 electrode disabled).

2. Toxic metabolic encephalopathies (such as a drug overdose) are associated with mixed alpha and beta frequencies that show some reactivity and typically have a favorable prognosis.
3. Posterior-predominant alpha may suggest a pontine lesion. Alpha coma of pontine origin should be differentiated from a locked-in syndrome where the EEG resembles normal wakefulness with reactivity to sensory stimulation and photic driving. Continuous recording in those with locked-in syndromes will also reveal evidence of state change including sleep architecture (rapid and non-rapid eye movement sleep). These may be fragmented or diminished in appearance due to disruption of the reticular networks. Prognosis is typically poor, with high mortality. Figure 23.6 shows alpha coma pattern in pontine injury [9].

Beta Coma

The EEG is characterized by diffuse 12- to 16-Hz beta activity with a frontal predominance. Reactivity to sensory stimulation is usually preserved in lighter forms. Sedative-hypnotic overdoses are the most common cause, usually with favorable prognosis. Figure 23.7 shows diffuse beta activity in the setting of benzodiazepine overdose [10].

CJD and SSPE

These are two infectious pathologies with distinct electrographic patterns.

The EEG in Creutzfeldt–Jakob disease (CJD) is characterized by frequent (every 1 s) generalized periodic discharges with a di- or triphasic morphology.

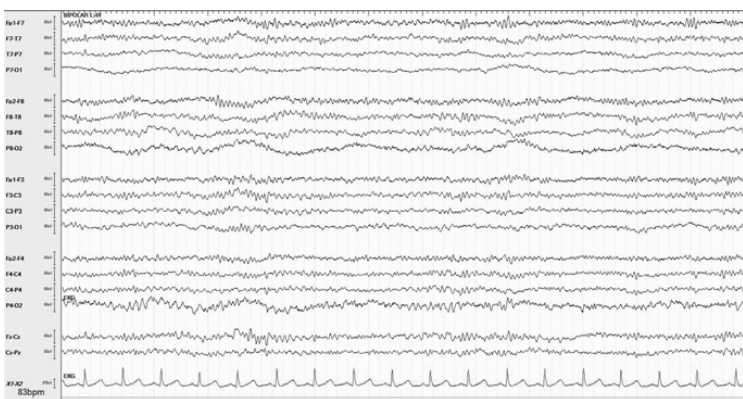


Figure 23.7 Twenty-eight-year-old woman with benzodiazepine overdose, diffuse excessive beta activity.

They may initially have a focal or lateralized predominance. The background is usually diffusely slow when the complexes first occur. Chapter 15, Figure 15.5 shows GPDs in CJD [11].

The EEG in subacute sclerosing panencephalitis (SSPE) is characterized by less frequent (every 4–14 s), high-amplitude, generalized periodic discharges (burst-like complexes) maximal in the frontocentral channels. Initially the background may be normal. The periodic complexes may be associated with myoclonic jerking (eyes or limbs), as shown in Figure 23.8 [12].

Extreme Delta Brush

Extreme delta brush is characterized by rhythmic (RDA) or periodic delta slowing with overriding bursts of rhythmic 20–30 Hz beta frequencies (plus fast features) that resemble beta-delta complexes in neonates. There is little reactivity or variability with sleep. This pattern, though nonspecific, has been described in many patients with anti-NMDA receptor encephalitis. Figure 23.8 shows extreme delta brush [13,14].

Chapter Summary

1. Global dysfunction (encephalopathy) results from impaired functioning of both cerebral hemispheres and/or brainstem.



Figure 23.8 Five-year-old boy with SSPE, generalized periodic discharges associated with myoclonic jerks.

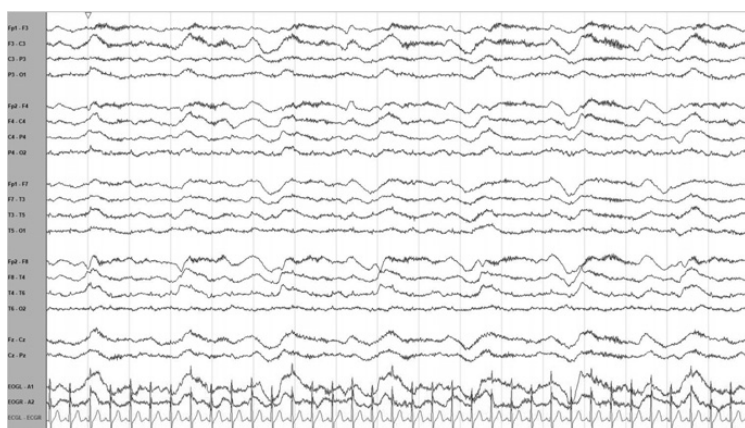


Figure 23.9 Same patient as in Figure 23.4, extreme delta brush pattern.

2. The EEG is an exquisitely sensitive tool to detect encephalopathy, but not specific to the underlying etiology.
3. The basic approach to an encephalopathic EEG consists of diagnosing encephalopathy, estimating its severity, and identifying repetitive patterns and NCSE, if present.
4. The three cardinal electrographic features of encephalopathy include background slowing, amplitude attenuation/suppression, and loss of reactivity.
5. Severe encephalopathies are typically characterized by a low-amplitude, slow and unreactive record, while a reversal of these trends may indicate improvement. Estimation of severity differs from prognostication.
6. Repetitive patterns (rhythmic and periodic) are common in encephalopathic patients and have important implications regarding etiology, epileptogenicity, and prognosis.
7. NCSE results from electrographic ictal activity that contributes to the encephalopathic state. It should be diagnosed based on clinical signs, EEG findings, and a response to antiepileptic medications. NCSE is independently associated with increased mortality.
8. Spindle coma is characterized by slow background with frequent symmetric spindles and typically has a favorable prognosis in those with reactivity and without evidence of structural damage.
9. Alpha coma consists of unreactive alpha frequencies; they have a posterior predominance in brainstem lesions and an anterior or diffuse distribution with cerebral anoxia. Posterior predominant alpha coma should be differentiated from a locked-in syndrome.
10. Beta coma typically occurs from drug overdose and usually has a favorable prognosis.
11. CJD and SSPE are infectious encephalopathies with distinct electrographic presentations that typically consist of GPDs.
12. Extreme delta brush pattern may suggest anti-NMDA receptor encephalitis.

References

1. Plum F, Posner JB. The diagnosis of stupor and coma. Oxford University Press, New York; 1982.
2. Kaplan PW. The EEG in metabolic encephalopathy and coma. *Journal of Clinical Neurophysiology*. 2004 Sep 1;**21**(5):307–18.
3. Hawkes CH, Bryan-Smyth L. The electroencephalogram in the “locked in” syndrome. *Neurology*. 1974 Nov 1;**24**(11):1015.
4. Sutter R, Stevens RD, Kaplan PW. Clinical and imaging correlates of EEG patterns in hospitalized patients with encephalopathy. *Journal of Neurology*. 2013 Apr 1;**260**(4):1087–98.

5. Young GB. The EEG in coma. *Journal of Clinical Neurophysiology*. 2000 Sep 1;17(5):473–85.
6. Kassab MY, Farooq MU, Diaz-Arrastia R, Van Ness PC. The clinical significance of EEG cyclic alternating pattern during coma. *Journal of Clinical Neurophysiology*. 2007 Dec 1;24(6):425–28.
7. Chong DJ, Hirsch LJ. Which EEG patterns warrant treatment in the critically ill? Reviewing the evidence for treatment of periodic epileptiform discharges and related patterns. *Journal of Clinical Neurophysiology*. 2005 Apr 1;22(2):79–91.
8. Kaplan PW, Genoud D, Ho TW, Jallon P. Clinical correlates and prognosis in early spindle coma. *Clinical Neurophysiology*. 2000 Apr 1;111(4):584–90.
9. Kaplan PW, Genoud D, Ho TW, Jallon P. Etiology, neurologic correlations, and prognosis in alpha coma. *Clinical Neurophysiology*. 1999 Feb 1;110(2):205–13.
10. Carroll WM, Mastiglia FL. Alpha and beta coma in drug intoxication. *British Medical Journal*. 1977 Dec 10;2(6101):1518.
11. Wieser HG, Schindler K, Zumsteg D. EEG in Creutzfeldt–Jakob disease. *Clinical Neurophysiology*. 2006 May 1;117(5):935–51.
12. Gökçil Z, Odabaşı Z, Aksu A, Vural O, Yardim M. Acute fulminant SSPE: clinical and EEG features. *Clinical Electroencephalography*. 1998 Jan;29(1):43–48.
13. Schmitt SE, Pargeon K, Frechette ES, et al. Extreme delta brush: a unique EEG pattern in adults with anti-NMDA receptor encephalitis. *Neurology*. 2012 Sep 11;79(11):1094–100.
14. Wang J, Wang K, Wu D, et al. Extreme delta brush guides to the diagnosis of anti-NMDAR encephalitis. *Journal of the Neurological Sciences*. 2015 Jun 15;353(1–2):81–83.